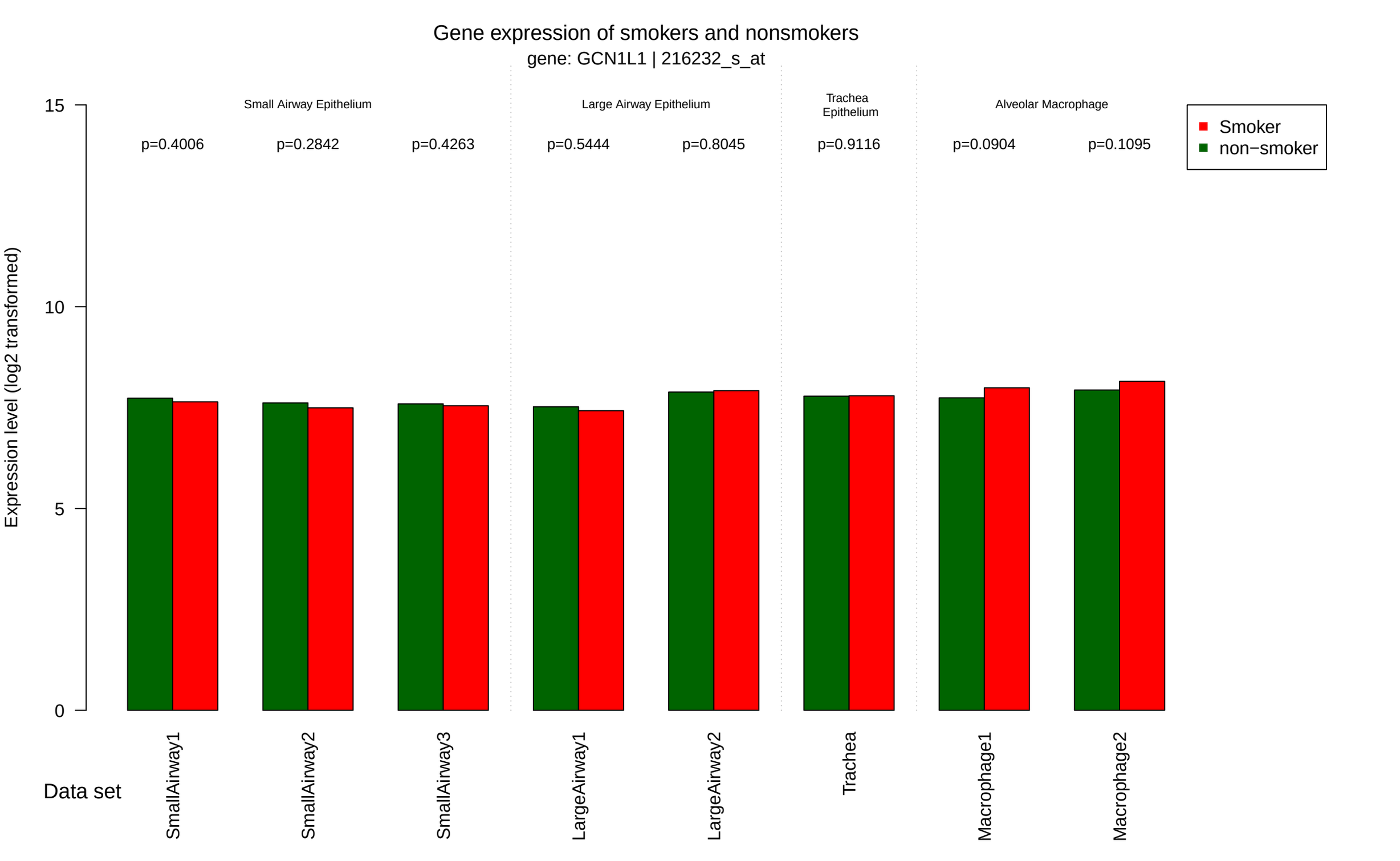
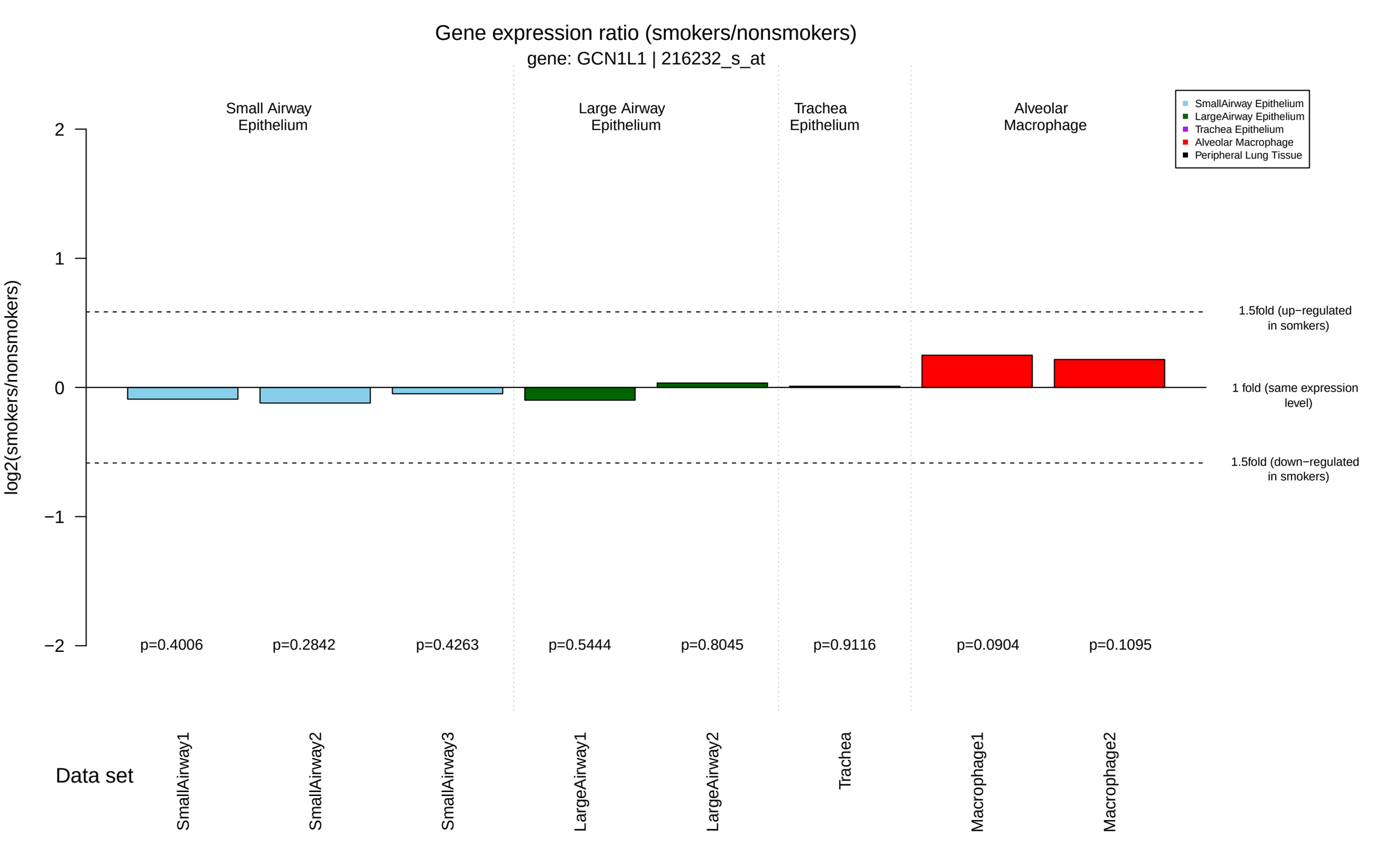
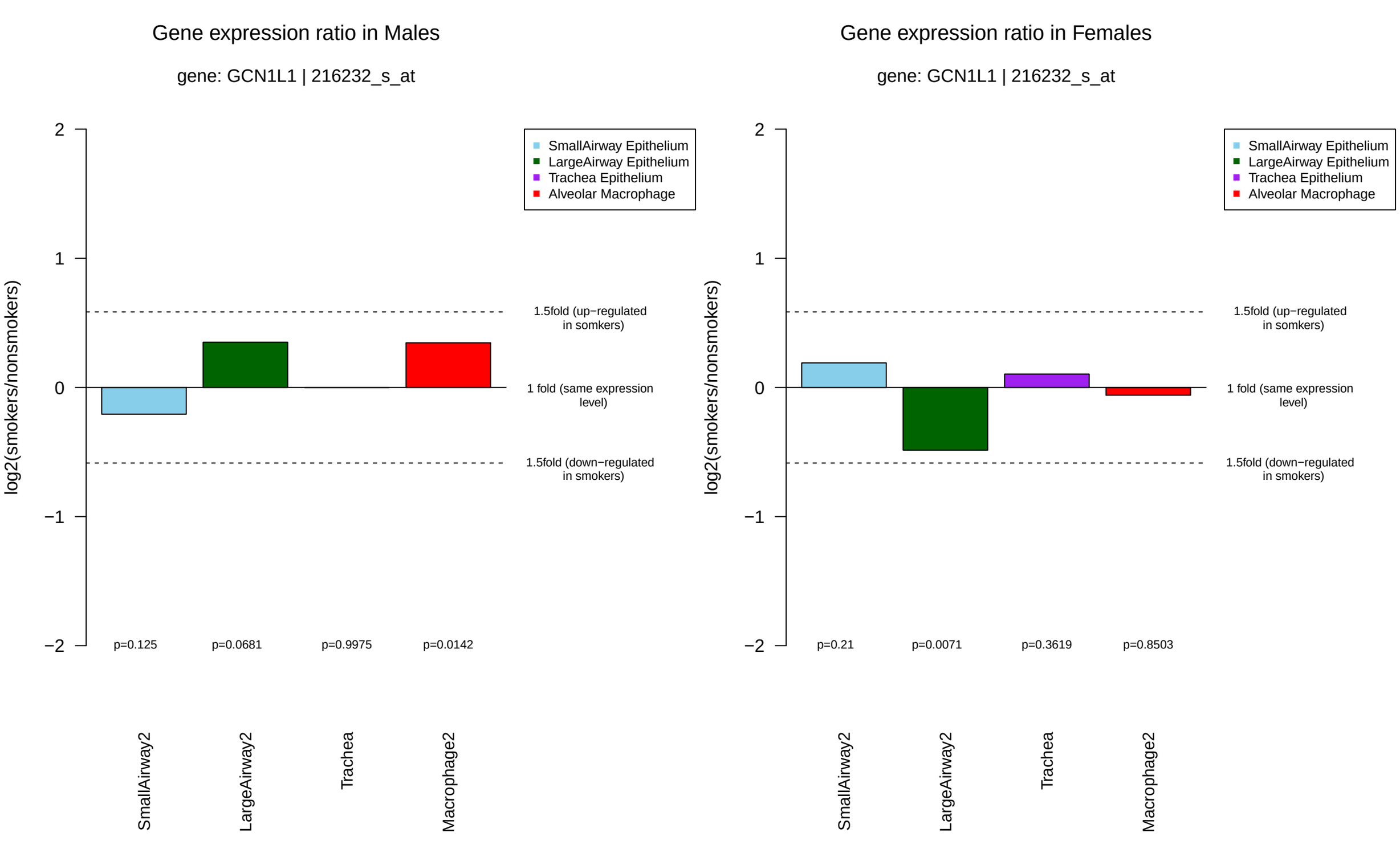
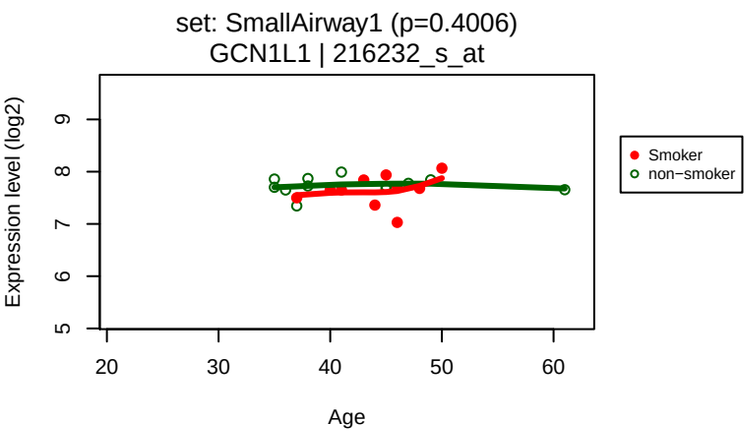
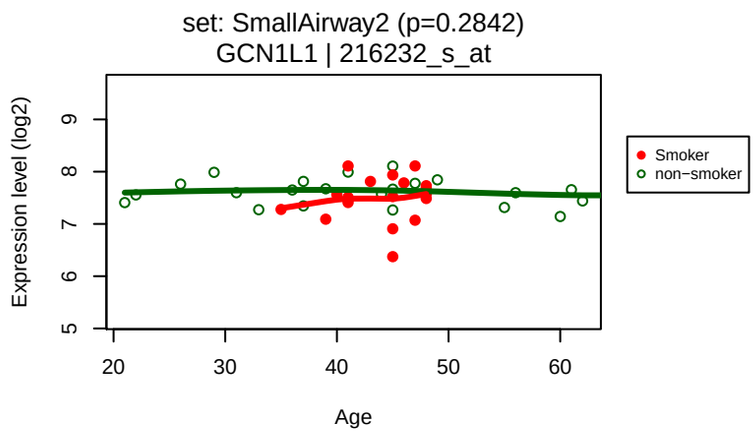
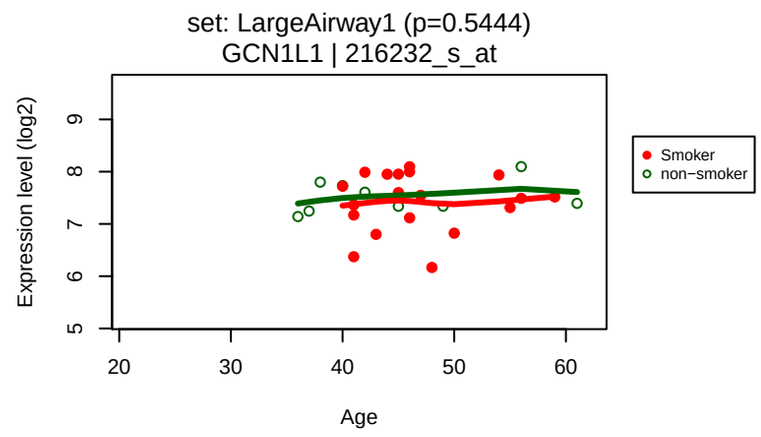

Gene expression of smokers and nonsmokers
gene: GCN1L1 | 216232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4006
p=0.2842
p=0.4263
p=0.5444
p=0.8045
p=0.9116
p=0.0904
p=0.1095
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

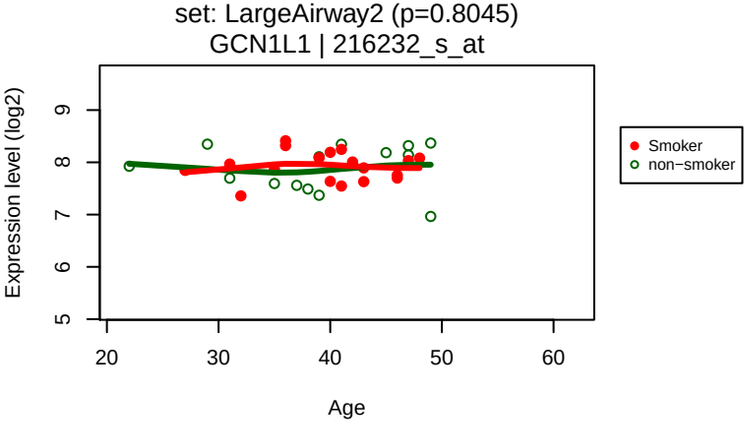

Gene expression ratio (smokers/nonsmokers)
gene: GCN1L1 | 216232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4006
p=0.2842
p=0.4263
p=0.5444
p=0.8045
p=0.9116
p=0.0904
p=0.1095
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

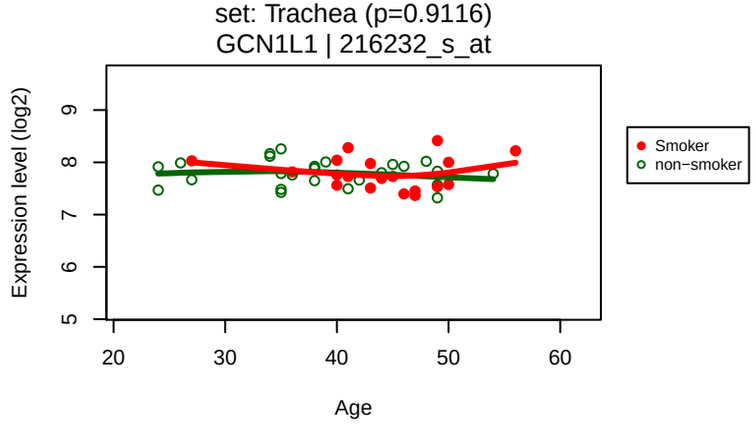

Gene expression ratio in Males
gene: GCN1L1 | 216232_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.125
p=0.0681
p=0.9975
p=0.0142
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCN1L1 | 216232_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.21
p=0.0071
p=0.3619
p=0.8503
SmallAirway2
LargeAirway2
Trachea
Macrophage2

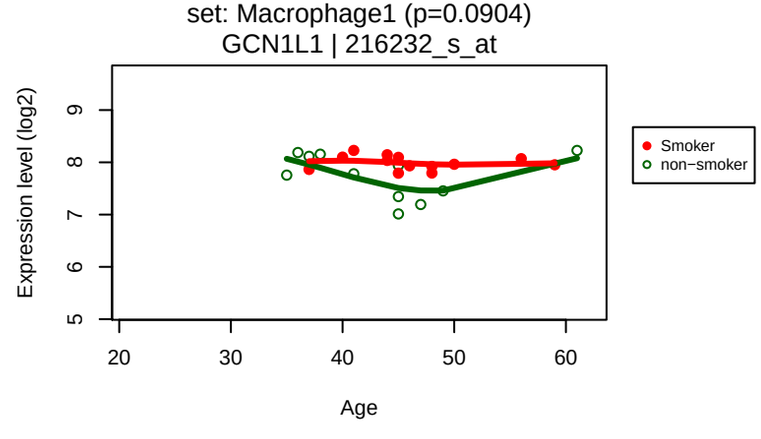

set: SmallAirway1 (p=0.4006)
GCN1L1 | 216232_s_at
Expression level (log2)
Age
Smoker
non−smoker

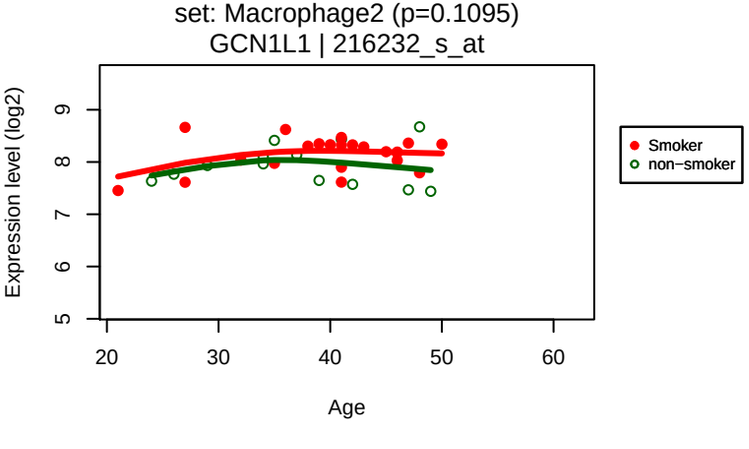

set: SmallAirway2 (p=0.2842)
GCN1L1 | 216232_s_at
Expression level (log2)
Age
Smoker
non−smoker

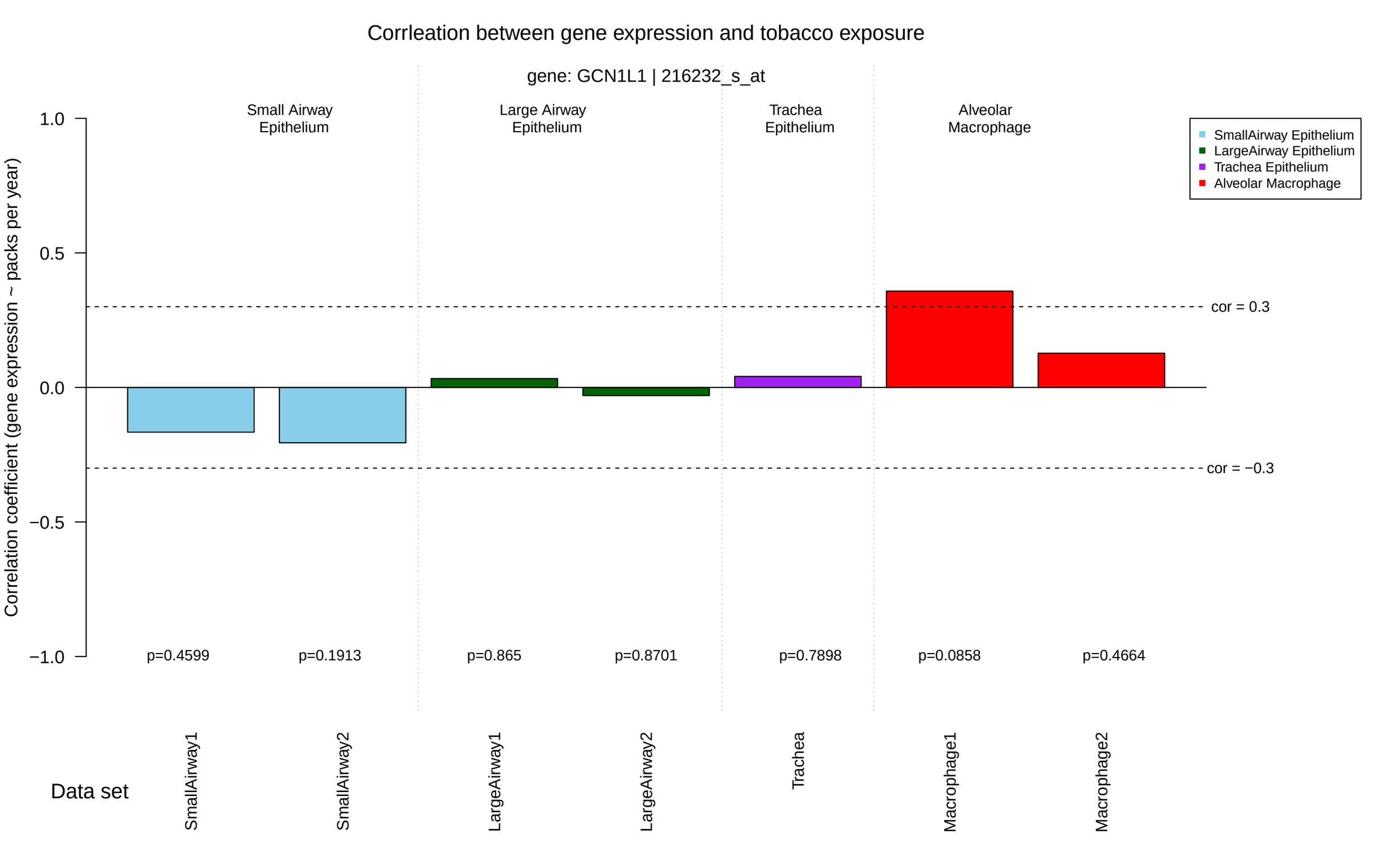

set: LargeAirway1 (p=0.5444)
GCN1L1 | 216232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8045)
GCN1L1 | 216232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9116)
GCN1L1 | 216232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0904)
GCN1L1 | 216232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1095)
GCN1L1 | 216232_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCN1L1 | 216232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4599
p=0.1913
p=0.865
p=0.8701
p=0.7898
p=0.0858
p=0.4664
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2